

phase : project design phase

Date	25 september 2026
Team ID	SWTID-2026-7377
project name	Script controlled ACL
Maximum marks	4

The Project Design Phase bridges the gap between initial requirements gathering and actual implementation/development. It translates system goals into detailed architectural, functional, and technical blueprints.

1. Key Objectives

- Define System Architecture: Establish high-level infrastructure, software design patterns, and data integration models.
- Detail Functional & Technical Requirements: Map out exact workflows, data models, UI wireframes, and business rules.
- Mitigate Technical Risk: Identify dependencies, security constraints, and performance bottlenecks prior to build.
- Establish Baselines: Lock down scope, design specifications, and test criteria for development teams.

2. Core Activities & Deliverables

Category	Key Activities	Output Deliverables
Architectural Design	System boundary planning, component modeling, technical stack selection, integration protocols (APIs).	System Architecture Document (SAD), Integration Maps
Data & Schema Design	Entity relationship mapping, database schema definition, field-level access controls, data dictionary.	ER Diagrams, Data Dictionary, Security Matrix

UI/UX Design	User journey mapping, wireframing, high-fidelity mockups, design system alignment.	Wireframes, Interactive Prototypes, UI Specs
Functional & Technical Logic	Process flow mapping, business logic definition, validation rules, user story writing.	Technical Design Document (TDD), Detailed Specs
Testing & Quality Strategy	Test case scenario definition, acceptance criteria mapping, performance benchmarking setup.	Test Plan, Traceability Matrix

3. High-Level Process Workflow

1. Input: Business Requirements Document (BRD) & Stakeholder Vision
2. Conceptual Design: Formulate high-level architecture options and evaluate trade-offs.
3. Detailed Design: Draft granular system models, database tables, component structures, and user flows.
4. Review & Validation: Conduct architecture reviews, security assessments, and UX walkthroughs.

5. Output & Sign-off: Finalize blueprints and hand off to development and QA teams.

4. Key Considerations & Guardrails

- Security & Compliance: Define role-based access control (RBAC), encryption methods, and compliance standards early.
- Scalability: Plan infrastructure capacity and performance targets to handle future load.
- Maintainability: Ensure code standards, modular designs, and documentation lower long-term technical debt.

Would you like to focus on a specific aspect of this phase, such as technical architecture templates, UI/UX workflows, or security design?